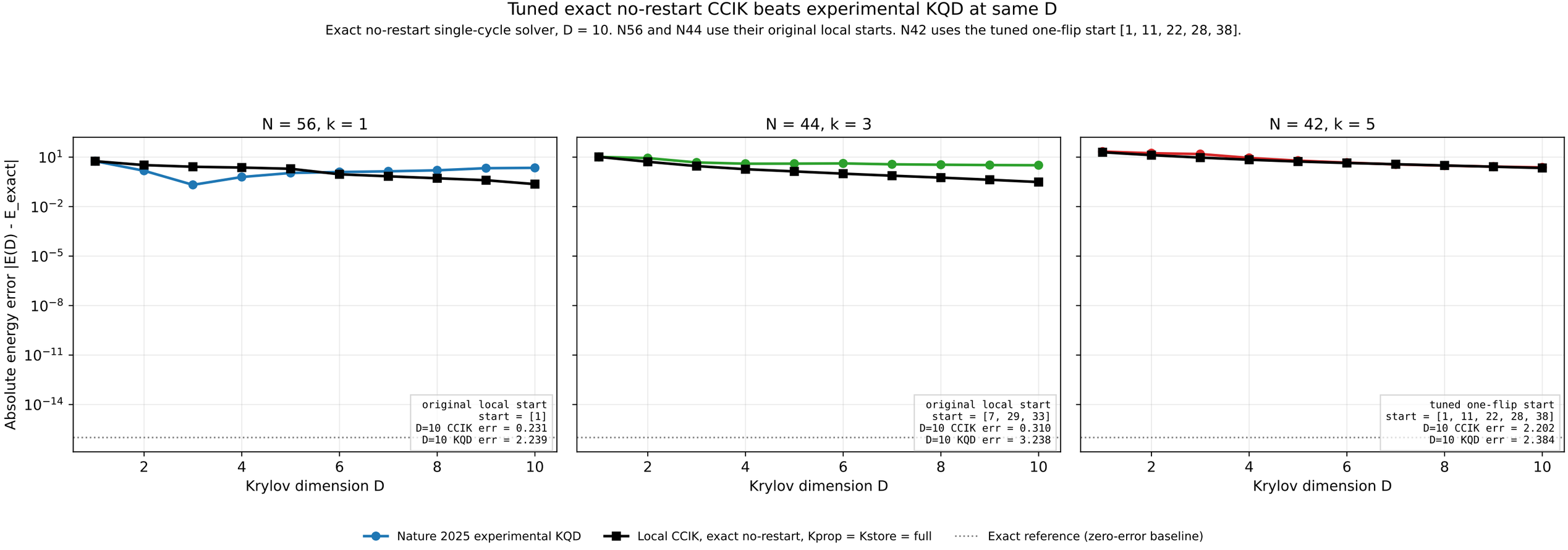
Tuned exact no-restart CCIK beats experimental KQD at same D
Exact no-restart single-cycle solver, D = 10. N56 and N44 use their original local starts. N42 uses the tuned one-flip start [1, 11, 22, 28, 38].
N = 56, k = 1
N = 44, k = 3
N = 42, k = 5
Absolute energy error |E(D) - E_exact|
Krylov dimension D
original local start
start = [1]
D=10 CCIK err = 0.231
D=10 KQD err = 2.239
original local start
start = [7, 29, 33]
D=10 CCIK err = 0.310
D=10 KQD err = 3.238
tuned one-flip start
start = [1, 11, 22, 28, 38]
D=10 CCIK err = 2.202
D=10 KQD err = 2.384
Nature 2025 experimental KQD
Local CCIK, exact no-restart, Kprop = Kstore = full
Exact reference (zero-error baseline)